Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Stefanescu, Doru M. Ph.D.		
Project Title:	Particle Engulfment and Pushing by Solidifying Interfaces (PEP)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	doru@coe.eng.ua.edu	**Fax:**	FY 256-348-8574
PI Organization Type:	UNIVERSITY	**Phone:**	256-348-1748
Organization Name:	University of Alabama, Tuscaloosa		
PI Address 1:	Dept. of Metallurgical and Materials Engineering		
PI Address 2:	141 Bevill Building, P.O. Box 870202		
PI Web Page:	http://www.matsceng.ohio-state.edu/		
City:	Tuscaloosa	**State:**	AL
Zip Code:	35487-0202	**Congressional District:**	7
Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	OTHER
Start Date:	02/01/1993	**End Date:**	09/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Sen, Subhayu	**Contact Phone:**	256 544-8264
Contact Email:	subhayu.sen@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Curreri, P. (NASA Marshall Space Flight Center) Dhindaw, B. (I.I.T. Kharagpur India) Sen, S. (Universities Space Research Association) Juretzko, F. (University of Alabama Tuscaloosa) Catalina, A. (Universities Space Research Association)		
Grant/Contract No.:	NAS8-00207		
Performance Goal No.:	1.25		
Performance Goal Text:	Use microgravity to establish and improve quantitative and predictive relationships between the structure, processing, and properties of materials.		

Task Description:	To acquire further insight in the physics of particle behavior, two complementing approaches are followed. The first is by developing a theoretical framework of the physics involved and subsequent computer modeling of the interaction between a particle and a solid/liquid interface. The second is by experimental validation. The models developed include a steady-state analytical model, followed by a dynamic model to account for the transient behavior when a particle at rest is approached by a solid/liquid interface. In order to account for convection currents and the influence of solutal accumulation for binary matrices a numerical model is currently under development. Partial experimental validation was done by directional solidification experiments. The first experiments used metallic matrices containing ceramic particles in a 1-g and micro-g environment. To actually observe the interaction of particles with a solidification front in in-situ, another set of experiments were carried out with transparent organic materials doped with insoluble particles in a 1-g and micro-g environment. Ground-based experiments under controlled solidification conditions with Al-SiC, Al-ZrO2, and Zn-ZrO2 systems to document pushing/engulfment phenomena in metallic systems were carried out. In addition, several tasks required in the flight requirements definition effort were performed. In addition to the original tasks, the Life and Microgravity Spacelab (LMS) mission flown between June 22 and July 6, 1996 provided the opportunity for some early (unscheduled) flight experimentation with a metallic system. Ground experiments were used to outline the processing parameters. Sample cartridges were designed and then tested with thermal models for compatibility with the sample design. The evaluation of the flight sample resulted in the determination of a critical velocity for the Al-ZrO2 system. It was found that this velocity is in the range of 0.5 to 1 μm/s which is in excellent agreement with the analytical model prediction of 0.775 μm/s. Another flight of opportunity was provided by the USMP-4 mission flown in November 1997. By using transparent organic materials the interaction of the particles with the solid/liquid interface was observed in-situ. They included the matrix-particle systems succinonitrile-polystyrene and biphenyl-glass. The particle size used for the samples ranged from 0.5 to 25 μm, resulting in a large number of data points. The results from these experiments were used to validate the analytical and numerical models. The experiments on the LMS mission were designed to achieve two goals. The first was to obtain preliminary data on metal/ceramic systems. The second goal was to evaluate the experimental approach for further experiments, currently scheduled for the International Space Station. The results from the LMS mission are invaluable for a number of crucial aspects of the experiment, such as selection of temperature profile, velocity range and proper design of the experimental cartridge necessary to obtain sound metallic samples. The primary objective of this task is to further develop the existing understanding on pushing and engulfment of particles by planar liquid/solid interfaces during solidification of metallic alloys. The approach towards achieving this objective is by developing an analytical and a numerical model describing the phenomenon, with a parallel experimental validation effort. The numerical model is required in order to include the effect of convection in the liquid on interface morphology and the particle/interface interaction. The interaction between an insoluble ceramic particle and a liquid-solid interface during solidification is significantly influenced by gravitational acceleration. Because of the influence of gravitational acceleration, microgravity experiments will be performed to validate these models and to further contribute to the experimental database for metals/ceramic particulate mixtures. It is of fundamental and practical importance to understand and control particle behavior at the solid/liquid interface during solidification. Particle behavior determines the uniformity of their distribution in the matrix. Uniformity of particle distribution is of great significance as it dictates the mechanical and physical properties of the composites. Introduction of insoluble ceramic particles in a metal matrix primarily involves three stages: transfer of particles from gas to liquid, interaction of particles in the liquid state, and finally, transfer of particles from liquid to solid. It is the last stage that is the most important and yet the least understood since it is the outcome of the interaction of numerous solidification variables. The anticipated results from this research program will provide a much better understanding of this stage of metal matrix composites processing and therein lies its significance. In addition to the interest for metal matrix processing, the problem is relevant to other areas including cryobiology (conservation of cells through freezing) and civil engineering (frost heaving in soil).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)